

Weak Identification Robust Methods for Production Function Estimation

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Motivation

Production functions are central to empirical work on **markups, TFP, and misallocation**

Standard approach: **control-function / proxy variable** methods following Olley and Pakes (1996); Levinsohn and Petrin (2003); Akerberg et al. (2015) (OP; LP; ACF)

Contribution: **Identification** underlying this method can be **weak**

- ▶ Point estimates are very **imprecise**
- ▶ **Non-normal** asymptotic distribution
- ▶ Standard Wald or Bootstrap confidence intervals **mislead**
- ▶ Goes undetected by conventional diagnostics — **potentially pervasive** in practice

This Paper

1. **Characterize weak identification** in the control-function framework

Two structural sources of near-rank failure

2. **Build an identification-robust toolkit** for OP/LP/ACF workflows

CUE; diagnostics; ID-robust inference

3. **Propagate uncertainty to markups**

Weak ID $\Rightarrow$ uninformative markup estimates

Related Literature

Identification & estimation of production functions

Control-function/proxy approach: Olley and Pakes (1996), Levinsohn and Petrin (2003), Akerberg et al. (2015); FOC approach: Gandhi et al. (2020); system GMM: Blundell and Bond (2000); survey: Akerberg et al. (2007)

Doraszelski and Li (2025): identification without “invertibility”

Weak identification in GMM

Theory: Staiger and Stock (1997), Stock and Wright (2000), Stock et al. (2002); robust inference: Kleibergen (2005), Cragg and Donald (1993), Angelini et al. (2024)

Documented in: NKPC (Kleibergen and Mavroeidis, 2009; Nason and Smith, 2008), monetary policy (Mavroeidis et al., 2014), DSGE (Canova and Sala, 2009; Iskrev, 2010; Andrews and Mikusheva, 2015), Euler equations (Yogo, 2004)

Markups and empirical applications

Markups & concentration: Hall (1988), De Loecker and Warzynski (2012), De Loecker et al. (2020), Autor et al. (2020); employer power: Yeh et al. (2022); Dobbelaere et al. (2024); misallocation: Gopinath et al. (2017); Caselli et al. (2021); Dobbelaere and Wiersma (2020)

Markup concerns: Raval (2023); Ridder et al. (2026)

Roadmap

1. Model and sources of weak identification
2. Estimation and robust inference toolkit
3. Monte Carlo evidence
4. Empirical application: Markups

Model and Control-Function Identification

Production function:

$$y_{it} = f(k_{it}, v_{it}; \beta) + \omega_{it} + \varepsilon_{it}, \quad \mathbb{E}[\varepsilon_{it} \mid k_{it}, v_{it}, \omega_{it}] = 0$$
$$\omega_{it} = g(\omega_{it-1}) + \xi_{it}, \quad \mathbb{E}[\xi_{it} \mid \omega_{it-1}] = 0$$

- ▶ k_{it} : predetermined (capital); v_{it} : static (labor or materials)
- ▶ ω_{it} : unobserved productivity

Control function:

$$\omega_{it} = h(p_{it}) + \eta_{it}, \quad \mathbb{E}[\eta_{it} \mid p_{it}] = 0$$

- ▶ Proxy $p_{it} = (k_{it}, \ell_{it}, m_{it})$. η_{it} : proxy noise (meas. error, frictions, ...)
- ▶ ID retained under index-sufficiency (Doraszelski and Li, 2025)

Two-stage ID

- ▶ **Stage 1:** $y_{it} = f(k_{it}, v_{it}; \beta) + h(p_{it}) + u_{1,it} \Rightarrow \mathbb{E}[y_{it} \mid p_{it}]$
- ▶ **Stage 2:** $y_{it} = f(k_{it}, v_{it}; \beta) + \tilde{g}(\mathbb{E}[y_{it-1} \mid p_{it-1}] - f(k_{it-1}, v_{it-1}; \beta); \rho) + u_{2,it}$
- ▶ **Moment conditions** $\mathbb{E}[\phi_{it}(\theta_0)] = 0$, with $\phi_{it}(\theta) = Z_{it} u_{it}(\theta)$

Two Structural Sources of Near-Rank Failure

Identification requires full column rank of the moment Jacobian $\mathcal{J} := \frac{\partial}{\partial \theta'} \mathbb{E}[\phi_{it}(\theta_0)]$.
When $\mathcal{J}$ is (near-)singular, the GMM criterion is (nearly) flat

Functional dependence between inputs

- ▶ Policy function of static inputs, e.g. labor, respond state variables: $v_{it} = v(k_{it}, \omega_{it})$
 - ▶ Controlling for ω_{it} , (nearly) functional dependence between v_{it} and k_{it}
- ⇒ Jacobian columns for β_v and β_k end up (nearly) collinear

Weak proxy

- ▶ Proxy p_{it} induces a noisy signal: measurement error, optimization frictions, heterogeneous input prices, or accounting aggregates
 - ▶ Relative to noise p_{it} gives no (little) explanatory power beyond k_{it}, v_{it}
- ⇒ Jacobian columns for β and ρ (nearly) collinear
- ▶ *Analogy to weak instruments: signal content of proxy plays the role of the first-stage coefficient in IV*

Estimation and Robust Inference Toolkit

ACF Estimator:

- ▶ Construct $\tilde{\omega}_{it}(\beta)$, concentrate out nuisance via OLS
- ▶ Optimize over concentrated moment conditions

CUE (Hansen et al., 1996) for prod. fct.:

- ▶ Optimal weighting of moments, exploiting full cross-moment covariance structure
- ▶ Required for ID-robust methods

Diagnostics

- ▶ **Cragg–Donald** (Cragg and Donald, 1993): Rank deficiency of Jacobian
- ▶ **Bootstrap normality** (Angelini et al., 2024): emp. distribution of bootstrap

Robust Confidence Sets

- ▶ **S-sets** (Stock and Wright, 2000): valid under *any* ID strength
- ▶ **K-sets** (Kleibergen, 2005): more powerful, orthogonal to misspecification

Monte Carlo

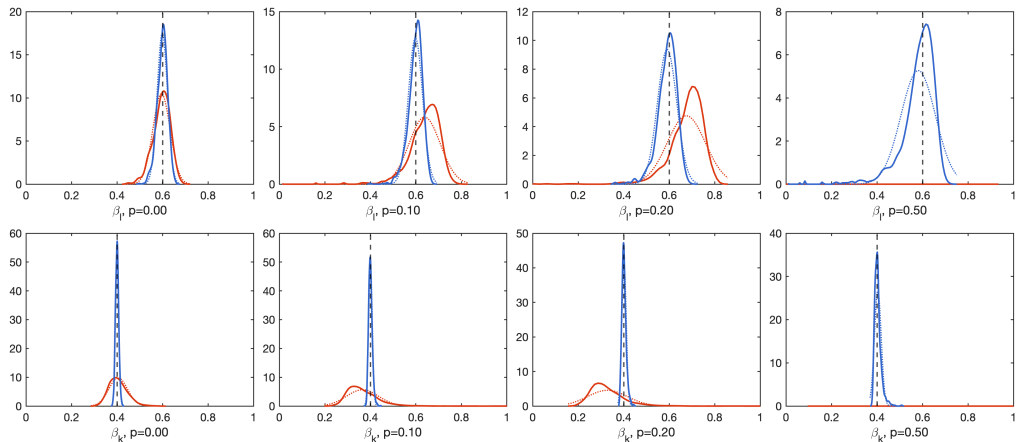
ACF framework: Firm optimization model with measurement error in materials

- ▶ Cobb–Douglas production: $(\beta_k, \beta_\ell) = (0.4, 0.6)$
- ▶ AR(1) productivity process: $\rho = 0.7$
- ▶ Labor optimization error ξ_ℓ to break functional dependence ($\sigma_{\xi_\ell} = 0.10$)
- ▶ Meas. err. in materials (proxy): 0%, 10%, 20%, 50% increase in $\text{Var}(m_{it})$

Findings:

- ▶ Sizeable meas. err. leads to weak ID
- ▶ Estimates ACF more unstable than CUE
- ▶ Diagnostics
 - ▶ Cragg–Donald no power against weak ID
 - ▶ Bootstrap normality test (Angelini et al., 2024) correctly sized and has power
- ▶ Inference
 - ▶ Nonrobust *Wald* test overrejects
 - ▶ Robust *S* and *K* tests size controlled

Monte Carlo: Estimates



ACF (red), CUE (blue). Solid = empirical density; dotted = normal fit. Columns: 0%, 10%, 20%, 50% meas. err. Top row: β_l ; bottom row: β_k .

Monte Carlo: Diagnostics and Inference

Meas. error	Diagnostics		$H_0: \theta = \theta_0$		
	CD	Normality	Wald	S	K
0%	1.000	0.057	0.064	0.060	0.060
10%	1.000	0.068	0.059	0.048	0.048
20%	1.000	0.167	0.075	0.062	0.062
50%	0.995	0.431	0.128	0.050	0.050

Rejection frequencies at 5% nominal level. CD: Cragg–Donald rank test (rejection = identification found). Normality: Doornik–Hansen bootstrap normality pre-test.

- ▶ **Cragg–Donald** rejects in nearly all replications — identification formally present
- ▶ **DH normality** rejection rate rises sharply — effective diagnostic for weak ID
- ▶ **Wald** over-rejects as proxy noise grows — conventional inference is unreliable
- ▶ **S and K** maintain correct 5% size throughout — robust inference works

Empirical Application

Production approach to markups

- ▶ Estimate production functions and derive firm-level markups: $\mu_{it} = \epsilon_{it}^v / s_{it}^v$ (De Loecker and Warzynski, 2012)
- ▶ Revenue data — uninformative about markup *levels* but informative about *dispersion* and *trends* (Ridder et al., 2026)

Data and model specification from Raval (2023)

- ▶ **Chile — ENIA** (plant-level survey): food products (ISIC 311), $\approx 16,000$ obs.; materials directly measured $\Rightarrow$ **high-quality proxy**
- ▶ **U.S. — Compustat** (firm-level accounting): food manufacturing (NAICS 31), $\approx 1,000$ obs.; materials mix costs and overhead $\Rightarrow$ **noisy proxy**

Findings

- ▶ Chilean survey data yields precise estimates (strong ID)
- ▶ Compustat data yields to highly imprecise estimates (Weak ID)
- ▶ Imprecision in output elasticities estimates propagated to markup estimates

Results: Chile (ENIA, ISIC 311)

Param.	ACF	CUE	95% CI		
			Wald	S	K
β_k	0.040	0.014	[0.006, 0.022]	[-0.011, 0.039]	[-0.011, 0.039]
β_ℓ	0.123	0.111	[0.087, 0.134]	[0.086, 0.136]	[0.086, 0.136]
β_m	0.905	0.947	[0.930, 0.964]	[0.922, 0.972]	[0.922, 0.972]
RTS	1.068	1.072	[1.060, 1.085]	[1.072, 1.097]	[1.072, 1.097]

$CD = 336.0$ ($p < 0.001$). $J_{ACF} = 40.116$ ($p < 0.001$); $J_{CUE} = 0.166$ ($p = 0.684$).
Bootstrap normality: all $p > 0.05$.

- ▶ Hansen test evaluated at ACF estimates rejects
- ▶ Cragg Donald rejects under-identification
- ▶ Bootstrap normality not rejected
- ▶ S and K sets narrow and close to Wald
- ▶ Strong identification is justified $\Rightarrow$ Wald inference reliable

Results: U.S. Compustat (NAICS 31)

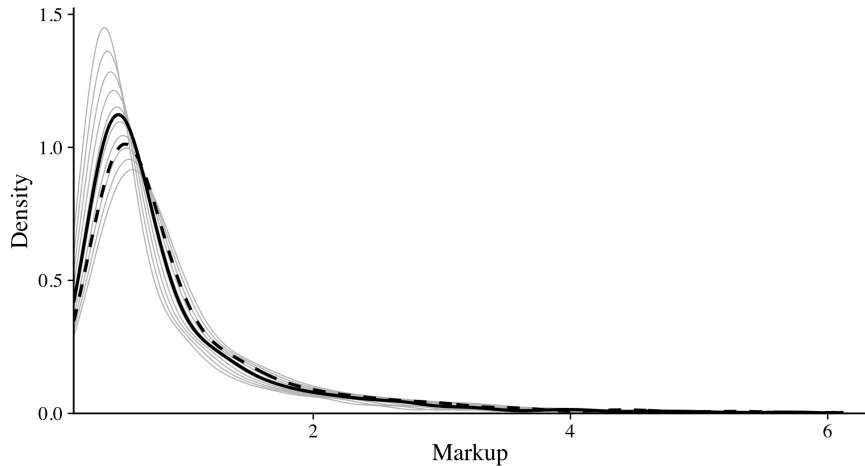
Param.	ACF	CUE	95% CI		
			Wald	S	K
β_k	0.329	1.697	$[-7.141, 10.534]$	$[\leq -3.3, 6.7>]$	$[\leq -3.3, 6.7>]$
β_ℓ	0.132	-0.992	$[-9.610, 7.626]$	$[\leq -6.0, 1.5]$	$[\leq -6.0, 4.0>]$
β_m	0.561	-0.538	$[-5.206, 4.129]$	$[\leq -5.5, 4.5>]$	$[\leq -5.5, 4.5>]$
RTS	1.022	0.166	$[-4.262, 4.595]$	$[-7.334, 2.666]$	$[-12.334, 10.166]$

$CD = 0.064$ ($p = 0.969$). $J_{ACF} = 12.026$ ($p = 0.001$); $J_{CUE} = 0.905$ ($p = 0.341$).

Bootstrap normality: all $p < 0.05$. " $\leq$ / $>$ " indicates set extends to grid boundary.

- ▶ Hansen test evaluated at ACF estimates rejects
- ▶ Cragg Donald does *not* reject under-identification
- ▶ Bootstrap normality rejected
- ▶ S and K sets unbounded (bounded for RTS)
- ▶ Points towards weak identification

Markups: Chile — Cross-Sectional Distribution 1996 (Labor)



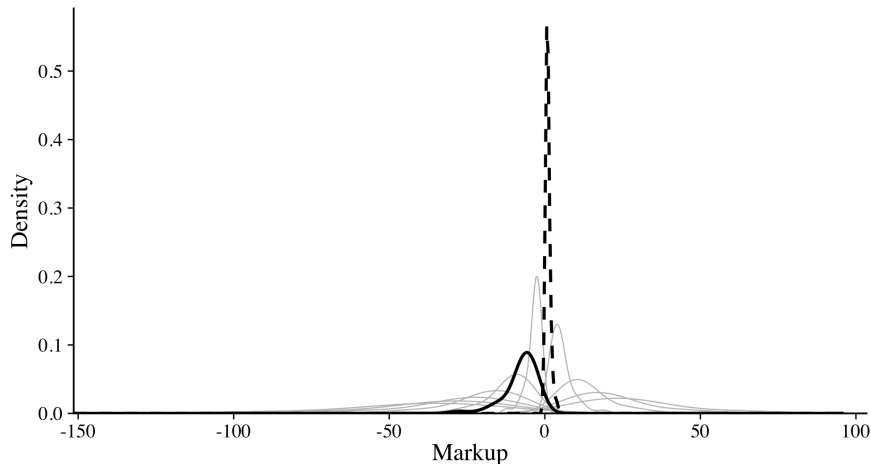
Markup distribution based on β_ℓ^{CUE} (solid), β_ℓ^{ACF} (dashed), $\beta_\ell \in [\underline{\beta}_\ell, \bar{\beta}_\ell]$ (light grey).

Markups: Chile — Time Series (Labor)



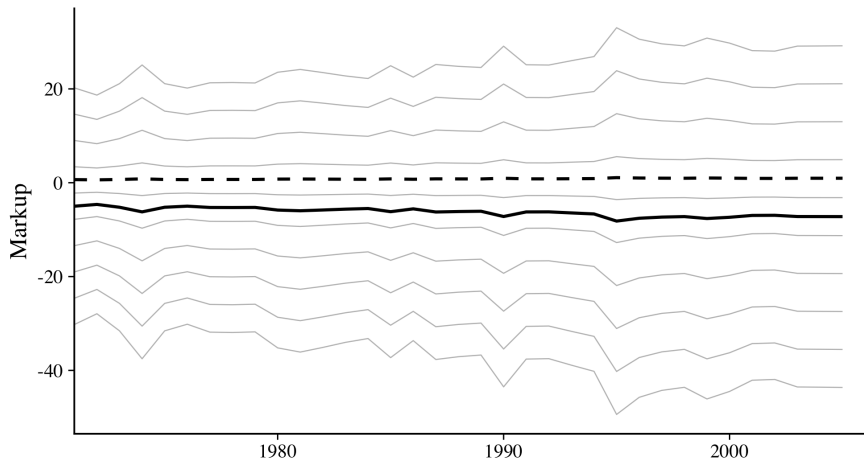
Markup time series based on β_ℓ^{CUE} (solid), β_ℓ^{ACF} (dashed), $\beta_\ell \in [\underline{\beta}_\ell, \overline{\beta}_\ell]$ (light grey).

Markups: U.S. — Cross-Sectional Distribution 1996 (Labor)



Markup distribution based on β_ℓ^{CUE} (solid), β_ℓ^{ACF} (dashed), $\beta_\ell \in [\underline{\beta}_\ell, \overline{\beta}_\ell]$ (light grey).

Markups: U.S. — Time Series (Labor)



Markup time series based on β_ℓ^{CUE} (solid), β_ℓ^{ACF} (dashed), $\beta_\ell \in [\underline{\beta}_\ell, \overline{\beta}_\ell]$ (light grey).

Conclusion

Sources of weak identification that are theoretically relevant

- ▶ Functional dependence between inputs
- ▶ weak proxies

Robust methods integrate directly into OP/LP/ACF workflows:

- ▶ Bootstrap normality diagnostic screens for weak-signal environments
- ▶ Identification-robust confidence sets (S, K) provide valid inference regardless of identification strength

Proxy quality matters in empirical settings

- ▶ researchers are encouraged to try alternative proxies
- ▶ use robust methods as a gauge

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